CLINICAL TEACHING NOTE | FAMILY MEDICINE

Chronic Hand Tremor

Diagnosis and management in middle-aged and elderly patients

1. Diagnostic approach

Step 1: Characterise the tremor (2018 IPMDS consensus)

- **Rest tremor:** limb fully supported, muscles relaxed.
- **Action tremor:** postural (arms outstretched), kinetic (finger-nose, pouring, writing), intention (worsens on approaching target), and task-specific (e.g. writing).
- Note **frequency** (rest 4-6 Hz, enhanced physiological 8-12 Hz), **distribution** (unilateral or bilateral, symmetric or not), and the **body parts involved** (head, voice, legs).

Step 2: Key history

- **Onset and tempo:** gradual over years suggests ET or PD; abrupt or stepwise suggests functional, drug-induced or structural causes.
- **Family history**, and response to alcohol (a typical ET feature).
- **Drugs:** sodium valproate, lithium, SSRIs/TCAs, salbutamol and other β-agonists, theophylline, amiodarone, corticosteroids, levothyroxine excess, antipsychotics, metoclopramide, prochlorperazine, cinnarizine/flunarizine, caffeine.
- Alcohol use or withdrawal, thyrotoxic symptoms, anxiety, weight loss, sweating.
- **PD-associated features:** hyposmia, constipation, REM sleep behaviour disorder, micrographia, reduced arm swing, slowness.
- Falls, imbalance, cognitive change, neuropathic symptoms.

Step 3: Examination

- Observe **at rest** (including while distracted, e.g. counting backwards), with **arms outstretched**, in the **wing-beating position**, and during **finger-nose testing**, spiral drawing and writing, pouring water, and drinking from a cup.
- Look for **bradykinesia** (finger tapping), **rigidity** and cogwheeling (with activation manoeuvre), gait, arm swing and postural stability.
- Look for dystonic posturing, a null point or sensory trick, head or voice tremor, cerebellar signs (dysmetria, nystagmus, ataxia), peripheral neuropathy, thyroid signs, and asterixis if liver or renal disease is suspected.
- **Functional pointers:** sudden onset, variable frequency or direction, distractibility, entrainment (tremor adopts the rhythm of a voluntary tapping task), and co-contraction.

2. Distinguishing the common causes

	Essential tremor	Parkinson's disease	Enhanced physiological	Cerebellar / intention
Type	Postural and kinetic	Rest (± re-emergent postural)	Postural, fine, fast	Kinetic, worsens toward target

	Essential tremor	Parkinson's disease	Enhanced physiological	Cerebellar / intention
Symmetry	Usually bilateral	Asymmetric at onset	Bilateral	Often unilateral or asymmetric
Other features	Head/voice tremor, family history, alcohol-responsive	Bradykinesia, rigidity, hyposmia	Reversible trigger (drug, thyroid, caffeine, anxiety)	Ataxia, nystagmus, dysarthria
Course	Slowly progressive	Progressive	Resolves with cause	Depends on cause (stroke, MS, alcohol, degenerative)

Other diagnoses to keep in mind

- **Drug-induced parkinsonism:** usually bilateral and symmetric.
- **Dystonic tremor:** irregular, position-dependent.
- **Orthostatic tremor:** leg tremor and unsteadiness on standing.
- **Neuropathic tremor** (e.g. demyelinating or paraprotein neuropathy).
- **Functional tremor.**
- **FXTAS:** an older male with tremor and ataxia.
- **Wilson's disease:** consider in anyone under about 40-50 with unexplained tremor or movement disorder.

> **Terminology (2018 classification)**
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> **ET plus:** essential tremor with additional soft neurological signs (e.g. mild dystonic posturing, impaired tandem gait).
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> **Isolated tremor syndrome of uncertain significance:** used when the picture doesn't fit a defined syndrome.

3. Investigations

Classic ET does not need extensive work-up. A reasonable primary-care set:

- **Bloods:** TSH/free T4, fasting glucose, FBC, U&E, LFTs, calcium, magnesium, B12.
- **Drug levels** if on lithium or valproate.
- **Serum ceruloplasmin** (± slit-lamp) if under about 40-50 or with atypical features.
- **MRI brain** if there are focal, cerebellar or asymmetric features, rapid progression, or a suspected structural cause.
- **DAT-SPECT** if ET versus PD or drug-induced parkinsonism remains unclear after specialist assessment.
- **Nerve conduction studies** or paraprotein screen if a neuropathic tremor is suspected.

Red flags: refer to neurology

- Abrupt onset.
- Unilateral tremor without a clear diagnosis.
- Cerebellar signs.
- Rapid progression.
- Cognitive, autonomic or psychiatric change.
- Onset after a stroke or head injury.
- A suspected functional disorder.
- Treatment failure.

4. Management

General measures for all causes

- **Explain the diagnosis.** Patients often fear Parkinson's; reassurance that ET is not PD (when the examination supports this) helps a great deal.
- Reduce caffeine, treat sleep deprivation and anxiety, and review and modify culprit drugs.
- **Occupational therapy:** weighted utensils and wrist weights, lidded or two-handled cups, larger-grip pens, and adaptive devices.
- Falls and safety review in the elderly.

Essential tremor

Treat only if the tremor causes functional or social disability. Doses are typical; check your formulary and local availability.

Line	Drug	Typical dose	Key cautions
First line	**Propranolol**	40-320 mg/day in divided doses, or long-acting form	Avoid or use cautiously in asthma/COPD, bradycardia or heart block, decompensated heart failure, and insulin-treated diabetes (masks hypoglycaemia). Elderly: postural hypotension, falls, fatigue, low mood.
First line	**Primidone**	Start 12.5-25 mg at night; titrate slowly, usually to no more than 250-750 mg/day	First-dose reaction (nausea, drowsiness, unsteadiness). Often poorly tolerated in the elderly: start low and go slow.
Second line	**Topiramate**	Slow titration	Cognitive slowing, paraesthesia, weight loss, renal stones.
Second line	**Gabapentin**	Slow titration	Sedation and dizziness; adjust for renal function.
Second line	**Clonazepam / alprazolam**	Low dose	Best avoided in the elderly (falls, dependence, cognitive effects).

- **Botulinum toxin injections** for refractory hand, head or voice tremor. Weakness is a risk, so hand injections need specialist skill.
- **Surgical options** for disabling, refractory tremor: thalamic (Vim) deep brain stimulation, MR-guided focused ultrasound thalamotomy (typically unilateral), or stereotactic radiosurgery. These need careful selection, with cognitive status and comorbidity weighed.

- **Alcohol** transiently helps, but recommending it is inappropriate because of dependence risk.

Parkinson's tremor

- **Refer to neurology.** Rest tremor is often less levodopa-responsive than bradykinesia and rigidity.
- Options include levodopa, dopamine agonists, MAO-B inhibitors and anticholinergics (avoid the last in the elderly because of cognitive effects).
- Physiotherapy and exercise are important adjuncts.

Enhanced physiological or drug-induced tremor

- **Treat or remove the cause:** control thyrotoxicosis, stop or reduce the drug, correct hypoglycaemia, and manage anxiety or alcohol withdrawal.
- Propranolol can be used situationally or short term.

Dystonic tremor

- Botulinum toxin is often the most effective treatment. Referral is usually appropriate.

Functional tremor

- Make a **positive diagnosis** and explain it clearly and non-dismissively.
- Physiotherapy or occupational therapy with a functional-movement focus, ± CBT, ideally with neurology and psychology input.

5. Follow-up

- Review response and side effects at **4-8 weeks** after starting or adjusting therapy.
- Re-examine periodically. Some patients diagnosed with ET later declare themselves as PD or dystonic tremor, so **revisit the diagnosis** if new features appear.

> **Key take-home points**
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> - Classify first: rest versus action, symmetry, frequency, associated signs.
> - Check drugs, thyroid and caffeine before labelling anything as ET or PD.
> - Treat ET only when it causes disability; propranolol and primidone are first line.
> - In the elderly, start low, go slow, and watch for falls.
> - Use the red flags to decide who needs neurology.

References

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This summary is an educational aid compiled from standard family medicine and movement-disorder references; it is not a substitute for local clinical guidelines or the primary literature.